

Wi-MoID: Human and Nonhuman Motion Discrimination Using WiFi With Edge Computing

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Abstract—Indoor intelligent perception systems have gained significant attention in recent years. However, accurately detecting human presence can be challenging in the presence of nonhuman subjects, such as pets, robots, and electrical appliances, limiting the practicality of these systems for widespread use. In this article, we propose a novel system (“Wi-MoID”) that passively and unobtrusively distinguishes moving human and various nonhuman subjects using a single pair of commodity WiFi transceivers, without requiring any device on the subjects or restricting their movements. Wi-MoID leverages a novel statistical electromagnetic wave theory-based multipath model to detect moving subjects, extracts physically and statistically explainable features of their motion, and accurately differentiates human and various nonhuman movements through walls, even in complex environments. In addition, Wi-MoID is suitable for edge devices, requiring minimal computing resources and storage, and is environment independent, making it easy to deploy in new environments with minimum effort. We evaluate the performance of Wi-MoID in five distinct buildings with various moving subjects, including pets, vacuum robots, humans, and fans, and the results demonstrate that it achieves 97.34% accuracy and 1.75% false alarm rate for identification of human and nonhuman motion, and 95.98% accuracy in unseen environments without model tuning, demonstrating its robustness for ubiquitous use.

Index Terms—Motion recognition, nonhuman motion identification, pet recognition, WiFi sensing.

I. INTRODUCTION

WITH the proliferation of Internet of Things (IoT) devices, indoor intelligent applications, such as security surveillance, intruder detection, occupancy monitoring, and activity recognition have gained significant attention [1]. However, these applications frequently suffer from an elevated rate of false alarms due to the inability to recognize human and nonhuman subjects, such as pets, robotic vacuum cleaners, and electrical appliances, as depicted in Fig. 1.

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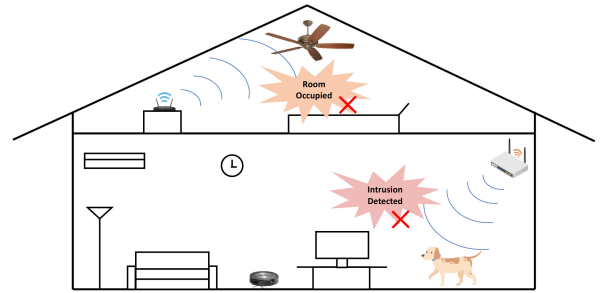


Fig. 1. Illustration of nonhuman motion interference with sensing systems.

Accurately discerning various moving subjects within smart environments is foundational for optimizing safety, energy efficiency, user interaction, and health monitoring. This differentiation bolsters safety by ensuring security systems accurately distinguish between, for example, human intruders and innocuous pet movements, thereby preventing false alarms. Efficient energy management is further realized when systems can discern a running appliance in a human’s absence, facilitating timely shutdowns. The nuanced identification of such moving subjects enables a tailored user experience, with IoT devices adjusting settings based on the detected entity, enhancing overall interaction. Moreover, for health and well-being monitoring, differentiating between a pet’s movement and a human’s is crucial to ensure precision in health insights. Given the prevalence of pets, robotic vacuum cleaners, and electrical appliances, especially in residential environments [2], it is crucial to develop a reliable system that can accurately recognize human and nonhuman subjects.

The accurate differentiation among human and various nonhuman movements still remains a challenge nowadays [3]. For instance, camera-based methods [4] and thermal-sensor-based approaches [5] can only detect moving subjects within the line of sight (LOS). Additionally, camera-based systems raise privacy issues [6]. Strategies leveraging radar to differentiate pets from humans based on vital signs expect pets to remain stationary, which is unrealistic [3], [7], [8]. More recently, WiFi-based intrusion detection systems have attempted to discern humans from pets by studying the effects on channel state information (CSI) [9]. However, these methods have strict device placement requirements and are often limited to LOS, or they presume the subject is moving within a predetermined area.

In this article, we introduce the first system (“WI-MOID”: WiFi-based human and nonhuman motion identification) that can accurately identify various human and nonhuman subjects through walls. The system utilizes ubiquitous WiFi signals and operates unobtrusively and without contact, eliminating the need for additional instrumentation or environmental restrictions. In contrast to existing systems that require pets, robots, and humans to move along predefined paths/directions/areas, WI-MOID can detect freely moving subjects, allowing them to walk/turn/stand/stay without any predefined restrictions. WI-MOID automatically detects the movement of the subject and extracts context-independent features, allowing it to function effectively in diverse environments without requiring additional training or parameter tuning.

However, realizing such a flexible system is not an easy task and several major challenges need to be addressed.

The first difficulty lies in deriving motion features in both LOS and Non-LOS (NLOS) settings using only a single-link WiFi signal. Speed pattern, a key component of motion features, presents its own unique challenge. While various WiFi-based speed estimation approaches, such as those using doppler frequency shift (DFS), have been attempted, they generally require multiple links or are only effective under LOS conditions. In this article, we draw inspiration from WiSpeed [10] and adopt a rich-scattering model to overcome the challenge by treating subjects as multiple scatters.

Second, developing a system that operates effectively in unseen contexts without requiring additional training or parameter adjustments is a substantial challenge. Most existing WiFi-based activity recognition methods rely on features extracted directly from raw CSI, which are susceptible to environmental changes with a degraded performance in new environments without retraining. To address this issue, we leverage a second-order statistic, the autocorrelation function (ACF), of the CSI and only extract features caused by the change of the multipath profiles. Unlike raw CSI, the ACF is insensitive to the static subjects and solely reflects the movement of moving subjects, making it well-suited for motion recognition in different environments.

Third, accurately recognizing pet motion in real-world scenarios is challenging, especially when their movement path, area, or activities are not restricted. Previous works that aim to recognize pet motion often make unrealistic assumptions about pets’ movement, such as assuming a specific path or confined area, considering only walking activity, or requiring pets to remain close to devices quietly for extended periods [9], [11]. In this article, we propose a new approach that combines physical and statistical features to classify human and nonhuman motion, and further enhance the classification using a hidden Markov model (HMM)-based state machine that leverages temporal information to boost the accuracy. By incorporating temporal information, WI-MOID can accurately and reliably distinguish pet motion from human motion, even when pets move freely and engage in different activities.

Furthermore, designing efficient and accurate algorithms for edge devices, such as smartphones and IoT systems, which

have less processing power and memory than conventional computers, is a challenge. Large deep learning models, while powerful, are often too resource intensive for these devices. Leveraging effective features with physical and statistical significance, WI-MOID addresses this with a lightweight support vector machine (SVM) model, which requires less computational power and fewer parameters, making it ideal for edge devices. Its minimal latency has been proven in real-world tests, affirming its real-time operation on devices with limited resources.

We implemented WI-MOID using commodity off-the-shelf (COTS) WiFi devices, and we have conducted extensive experiments in five typical indoor environments: two apartment; a townhouse; a single-family house; and a typical office space. Most of the data were obtained in NLOS conditions to ensure real-world relevance. WI-MOID demonstrates impressive performance with a validation accuracy of 97.34% and a testing accuracy of 95.98% in unseen environments. The high accuracy, coupled with the environment-independent features, positions WI-MOID as a promising solution for detecting, differentiating, and monitoring human and nonhuman motion across diverse and pervasive contexts. It offers potential applications in areas, such as security, home automation, and activity recognition.

In summary, our contributions in this work are as follows.

- 1) We have developed a motion detection and recognition system capable of accurately monitoring and recognizing human and nonhuman subjects. By extracting a wide spectrum of physically plausible and statistical features, WI-MOID can detect and recognize humans, pets, robots, and other electrical appliances, regardless of their position, orientation, and environmental circumstances.
- 2) We have designed an HMM-based state machine that incorporates temporal information to correct misclassifications, enhancing the system’s capability to accurately recognize complex motion and improving its overall accuracy and stability in practical applications.
- 3) We developed a motion classification scheme that identifies human and nonhuman subjects without relying on data-intensive deep learning methods, and compiled a comprehensive CSI data set of human motion and various nonhuman motions.
- 4) We developed a prototype of WI-MOID using readily available COTS WiFi devices and conducted experiments in diverse indoor environments. The results demonstrate high accuracy and suitability for deployment in various settings. Furthermore, we have implemented WI-MOID to run in real time on various edge devices with limited computing resources and storage, indicating its readiness for ubiquitous deployment.

The structure of this article is organized as follows: Section II discusses related literature. Section IV delves into the details of motion detection and speed estimation. Section V introduces feature extraction and motion recognition. Experimental setups and evaluations are described in Section VI, and we draw our conclusions in Section VIII.

II. RELATED WORKS

While there have been numerous indoor activity recognition and monitoring methods proposed over the years, only a few have taken into account the impact of nonhuman motion. Some researchers have attempted to filter out pet motion from human motion due to its potential interference with their systems. In this section, we review the current methods used to differentiate pets from humans, focusing on non-RF-based and RF-based methods.

A. Non-RF-Based Methods

Non-RF-based methods distinguishing pet motion and human motion mainly rely on heat maps obtained from thermal sensors [5] or image/face recognition techniques using cameras [4]. These methods have limited coverage, and are often restricted to LOS, and camera-based methods further introduce privacy concerns [6].

B. RF-Based Methods

Recent attempts to differentiate between humans and pets have employed radar [8] and ultrawideband (UWB) [12], focusing on analyzing biometric indicators, such as heartbeat and respiration. But these methods necessitate the pet to remain stationary during recognition and impose constraints regarding their spatial position and distance from the device, thereby rendering them impractical and ineffective for identifying moving subjects. While [7] enables the recognition of canine movement distinct from human movement through radar, it demands that the device be installed on the ceiling, and restricts the movement of humans and pets to a limited area under the device. Furthermore, the method presented in [13] can differentiate fan motion from human respiration, yet it remains inapplicable to human activities, such as walking and running.

WiFi-based sensing systems are attractive for indoor activity recognition and monitoring due to their ubiquity and low cost [14], [15]. Numerous WiFi-based systems have been designed to monitor human motion and the environment. WiFi-based motion detection systems [16], [17] detect the occurrence of human motion within an environment, while WiFi-based activity recognition methods [18], [19], [20], [21], [22] attempt to recognize human daily activities, such as running, walking, and sitting. Identification systems, such as [23], [24], [25], and [26] recognize human identity through gait or other biometric features extracted from WiFi signals. In addition, mapping and tracking systems that leverage WiFi [27], [28], [29], [30] plot human movement paths and generate floor maps. Nonetheless, these systems fail to account for the influence of nonhuman entities, such as pets and vacuum robots, on their operational accuracy. These nonhuman subjects, capable of affecting WiFi signals similarly to humans, introduce a potential source of interference. Consequently, these systems' reliability for indoor monitoring and recognition can be compromised, particularly when nonhuman subjects are present in the environment.

Some researchers have proposed to detect intrusion using WiFi. Among the proposed systems, PerFree [9] and M-WiFi [11] proposed methods to reduce false alarms caused by pets. For example, PetFree distinguishes pets from humans by mapping the subject's effective interference height (EIH) to CSI measurements. This method, however, requires the transmitter and receiver to be positioned at the same height and it operates solely under LOS conditions. Moreover, its performance can be affected by the multipath effect in environments with complex layouts. M-WiFi differentiates between pets and humans by examining the RF disturbance. However, this system encounters issues when a pet moves close to the wireless link. This movement can induce signal fading akin to that of a person moving farther from the wireless link, resulting in a significant number of false alarms, regardless of differences in RF disturbance.

A preliminary version of the system [31] focused on the motion classification of human and nonhuman subjects without differentiating between the various nonhuman subjects. In this article, we expand its scope by further classifying nonhuman subjects and propose a novel state machine based on the Hidden Markov Chain to take into account historical information and temporal characteristics of the outputs, enabling more accurate and stable recognition of moving subjects in the environment. We evaluate the performance on different sounding rates of CSI, demonstrating its effectiveness in the range of 30 to 1500 Hz. Furthermore, we evaluate the computational resources required for the real-time operation of WI-MOID and prove that WI-MOID can run in real time on commercially available edge devices.

III. SYSTEM OVERVIEW

A. Insight

A moving subject can be categorized as either human or nonhuman, the latter encompassing a range of entities, such as pets or vacuum robots. When we refer to human motion, we are talking about the movement of the human body, encompassing activities, such as walking, running, and sneaking. In contrast, nonhuman motion pertains to the movement of animate or inanimate subjects, which includes animals, robots, and other physical objects. We classify any motion originating from nonhuman entities as nonhuman motion.

Interestingly, though pets, cleaning robots, and even electrical appliances such as fans can all generate motion that can be picked up by sensing systems, their movements are distinct from those of humans. The differences in gait patterns are a key aspect of this distinction, with humans, pets, and robots each demonstrating unique movement patterns due to their bipedal, quadrupedal, and wheeled locomotion, respectively. These differences are visually represented in Fig. 2. As a result of these disparate movement patterns, the analysis of features like speed patterns becomes an essential factor in differentiating the motion of these subjects.

B. System Overview

The overview of WI-MOID, as illustrated in Fig. 3, is primarily composed of the following three steps.

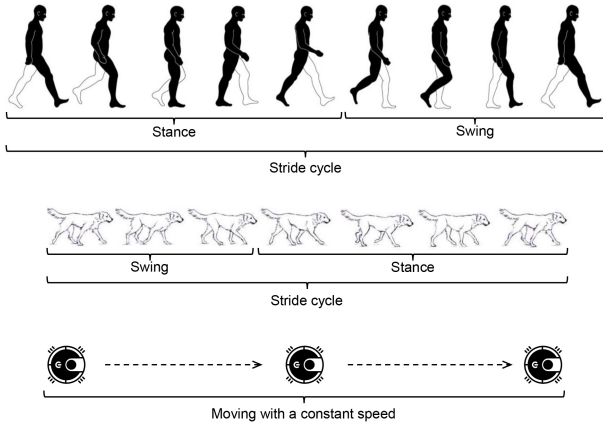


Fig. 2. Illustration of moving patterns of human, dog, and cleaning robot.

- 1) We preprocess the CSI received by the receiver. We compute the ACF of the CSI and, based on this, calculate the motion statistic to determine the presence of moving targets in the environment. Additionally, we derive the speed of the moving target based on the ACF.
- 2) We effectively extract a series of features from the moving target. Using the ACF, motion statistic, and speed, we extract physically interpretable features and signal statistical features of the moving target. These features are independent of the environment, location, and direction.
- 3) Finally, using SVM and a state machine, we output the final identification result of the system. We initially use SVM to classify the features of the target and then employ a state machine to adjust any potential misclassifications based on the transitional features between states, finally outputting the result.

IV. MOTION DETECTION AND SPEED ESTIMATION

This section presents our approach to detecting the motion of subjects and estimating their speeds through walls, using a single WiFi link.

A. Preliminary

For wireless communication systems with omnidirectional antennas, the WiFi signal undergoes multiple reflections as it propagates through the indoor environment, resulting from the presence of various reflecting surfaces. The CSI of a WiFi signal provides information about the wireless channel between the transmitter and the receiver. The CSI of the multipath channel at a given time t and frequency f can be expressed as follows:

$$H(t, f) = \sum_{l=1}^L a_l(t) \exp(-j2\pi f \tau_l(t)) \quad (1)$$

where $a_l(t)$ and $\tau_l(t)$ denote the complex amplitude and propagation delay of the l th multipath component (MPC), respectively, and L stands for the number of MPCs contributing to the CSI.

In order to accurately detect the motion of a subject in an indoor environment, it is crucial to consider the presence of numerous multipath signals arising from rich scattering. A statistical model that account for all these multipath components can be employed to estimate the speed of the target [10], [16], [23].

In indoor environments, rich scattering occurs due to both static and dynamic scatterers. Static scatterers comprise walls, floors, and stationary furniture, while dynamic scatterers include moving individuals or objects. The superposition principle of electromagnetic (EM) waves allows us to decompose CSI as follows:

$$H(t, f) = \sum_{i \in \Omega_s(t)} H_i(t, f) + \sum_{j \in \Omega_d(t)} H_j(t, f) + \varepsilon(t, f) \quad (2)$$

where $\Omega_s(t)$ represents the set of static scatterers, $\Omega_d(t)$ corresponds to the set of dynamic scatterers. $H_i(t, f)$ and $H_j(t, f)$ represent the contributions of the i th and j th scatterers, respectively. The noise term, $\varepsilon(t, f)$, is statistically independent of $H_i(t, f)$ and $H_j(t, f)$ [10]. Each scatterer acts as a “virtual transmitter,” scattering its received EM waves around. The CSI represents the aggregate of the electric fields of all incoming EM waves.

The ACF of the CSI $H(t, f)$ with a time lag τ , denoted as $\rho_H(\tau, f)$, can be determined by calculating the covariance between $H(t, f)$ and $H(t + \tau, f)$ divided by the variance of $H(t, f)$ itself. Mathematically, it can be expressed as follows:

$$\begin{aligned} \rho_H(\tau, f) &= \frac{\text{Cov}[H(t, f), H(t + \tau, f)]}{\text{Cov}[H(t, f), H(t, f)]} \\ &= \frac{\sum_{i \in \Omega_d} \sigma_{F_i}^2(f) J_0(kv_i \tau) + \sigma^2(f) \delta(\tau)}{\sum_{i \in \Omega_d} \sigma_{F_i}^2(f) + \sigma^2(f)}. \end{aligned} \quad (3)$$

Here, $J_0(\cdot)$ is the Bessel function of the first kind, given by $J_0(x) = (1/2\pi) \int_0^{2\pi} \exp(-jx \cos(\theta)) d\theta$. $F_i(\cdot)$ denotes the channel gain of the MPC for the i th scatter, and $k = (2\pi/\lambda)$ is the wave number where λ is the wavelength. The term $\delta(\cdot)$ represents the Dirac delta function.

B. Motion Detection and Speed Estimation

The motion statistic $\phi(f)$ for a subcarrier with frequency f is defined as ACF of the CSI $H(t, f)$ with a time lag of $\tau = 1/F_s$, where F_s is the sounding rate [16]. That is

$$\phi(f) \triangleq \rho_H\left(\tau = \frac{1}{F_s}, f\right). \quad (4)$$

Motion statistics function as a reliable gauge of movement presence or lack thereof within a given environment. In a stationary environment, the motion statistic $\phi(f)$ is close to 0, whereas in dynamic environments with movement, $\phi(f) > 0$. Fig. 4 illustrates the motion statistics of human, pet, and robot movements.

Despite the discrepancies in size, various subjects are capable of producing similar disruptions across different distances. This is evident from the analogous motion statistics exhibited in Figs. 4 and 5(a). Consequently, relying solely on motion statistics cannot produce reliable recognition results.

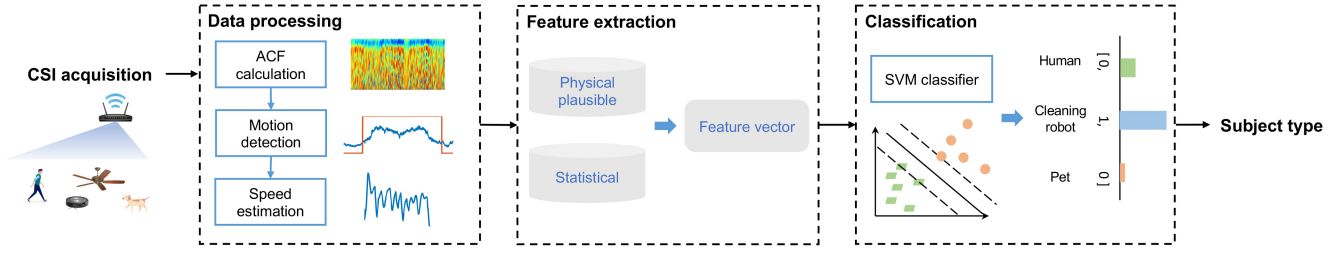


Fig. 3. System overview.

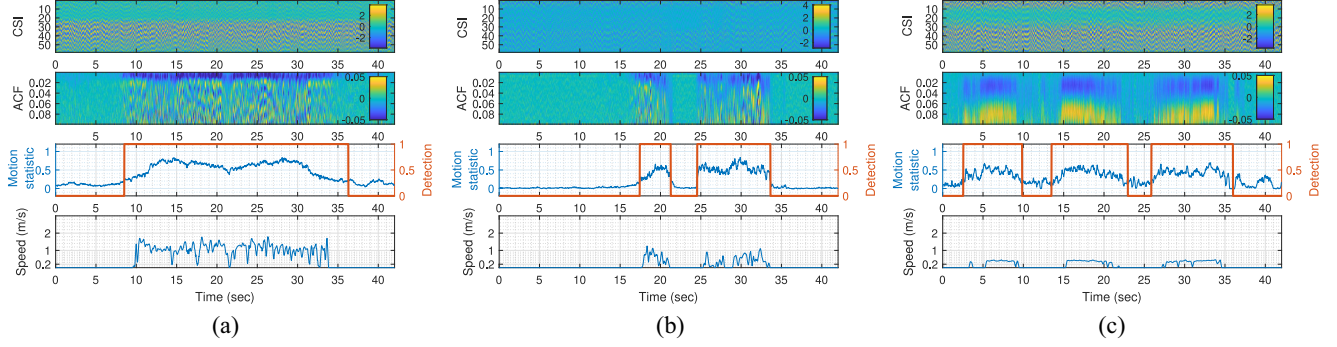


Fig. 4. CSI, ACF, motion statistics, and speed estimation for (a) human, (b) pet, and (c) cleaning robot.

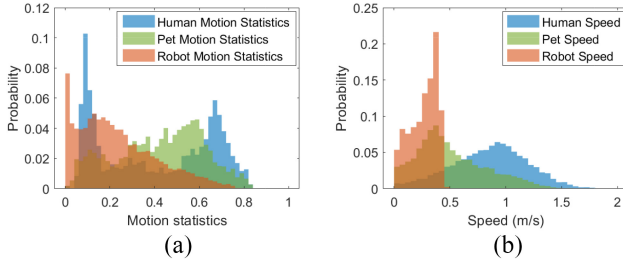


Fig. 5. Histograms of (a) motion statistics and (b) speed estimations for human and pet.

Considering that speed can effectively capture the distinct walking patterns exhibited by humans and nonhuman entities, we extend our feature set by extracting speed information from WiFi signals.

To approximate the motion speed of a target, we simplify the ACF of the CSI by employing a motion model that assumes the torso contributes most of the strong scatterers, resulting in $\rho_H(\tau, f) = \alpha(f)J_0(kv\tau)$ [23], where $\alpha(f)$ represents the gain of each subcarrier. This function bears resemblance to the well-known Bessel function $J_0(x)$, where $x = kv\tau$. By aligning the position of the first peak or valley of $J_0(kv\tau)$ with that of the Bessel function, we can estimate the subject's speed. Specifically, we detect the first peak and calculate the speed as $\hat{v} = (x_0/k\hat{\tau}) = (x_0\lambda/2\pi\hat{\tau})$, where x_0 denotes the constant value corresponding to the first peak of $J_0(x)$, and $\hat{\tau}$ represents the time lag corresponding to the first peak in $J_0(kv\tau)$. Estimated speeds of human, pet, and robot motions derived from the ACF are depicted in Fig. 4.

Although estimating the speed of a moving subject provides significant insights for subject differentiation, it is inadequate to rely solely on speed values to distinguish between

human and nonhuman subjects, due to the potential overlap in their speed values, as shown in Fig. 5. This is because their movements may exhibit significant overlap in terms of speed values. Therefore, a comprehensive analysis of motion statistics, speed, and ACF patterns over time is imperative for the successful distinction between moving subjects. By extracting features that have physical and statistical explanations from these patterns, we can enhance discrimination and accurately identify human and nonhuman subjects. The subsequent section will elaborate on these features and their role in motion classification.

V. FEATURE EXTRACTION AND MOTION RECOGNITION

This section outlines the methodology for distinguishing between the motion of human, pet, and cleaning robot. First, a set of features is extracted, followed by the development of a recognition model that utilizes the selected features.

A. Feature Extraction

In our pursuit to discern features that can effectively identify movements of human and nonhuman subjects, we adopt a two-pronged approach: 1) examination of physical motion characteristics and 2) statistical properties gleaned from CSI. On the physical front, we delve into gait-related attributes and speed patterns exhibited by the subjects. Statistically, we scrutinize the statistical features of ACF and changes in motion statistics. This combined approach aims to encapsulate the holistic nature of movement and its impact on the WiFi signal, facilitating accurate identification across varied subjects.

1) *Physical Features Extraction*: The term “gait” signifies a pattern of limb movement during locomotion on a solid surface and has proven an effective human identification feature. Humans walk bipedally, pets, such as dogs and cats

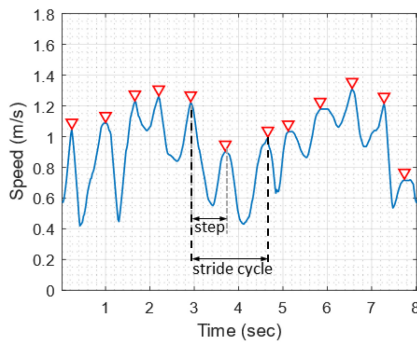


Fig. 6. Example of human stride cycle.

predominantly operate as quadrupeds, and cleaning robots rely on the wheel-based motion. These inherent differences result in unique gait and speed patterns, providing differentiation between humans, pets, and robots.

In locomotion, the movement pattern of animals' limbs generates the gait, where humans and pets alternate their feet. A stride cycle encompasses a stance phase (foot-ground contact) and a swing phase (foot lift and forward motion). Speed changes are integral to gait, decreasing during the stance phase and increasing during the swing phase, as shown in Fig. 6. In contrast to humans and pets, vacuum robots do not exhibit significant speed fluctuations, enabling their differentiation via gait detection. We estimate motion speed using the method outlined in Section IV, identifying patterns of speed through peak and valley analysis. Gait presence was determined by detecting speed fluctuations with a distinct ups and downs regularity.

Stride Cycle Time and Stride Length: Stride cycle time and stride length are essential gait indicators. Stride cycle time refers to the duration between the same foot contacting the ground twice, while stride length is the distance covered by a person or animal between two consecutive footfalls. These definitions are applicable to quadruped pets, such as dogs and cats. Since pets generally have shorter legs than humans, their stride lengths are smaller than that of humans during normal indoor movement.

We extract these two features from the speed curves to distinguish the movement of human and pet movements. Specifically, the stride cycle time is extracted by calculating the time duration of three consecutive speed peaks. Stride length is the integral of speed value over stride cycle time. Fig. 7 illustrates the average stride lengths of human and dog across ten motion instances. It can be observed that although humans and pets may have similar stride cycle times, their stride lengths are significantly different. Pets exhibit much shorter stride lengths compared to humans due to their shorter legs.

Speed Mean and Deviation: As wheeled platforms lack feet, vacuum robots typically maintain a constant speed while moving from one location to another and slow down when encountering obstacles. Therefore, speed-based features are analyzed to differentiate between the movements of vacuum robots and humans. The average speed is calculated as the mean of the instantaneous speeds of a moving target during

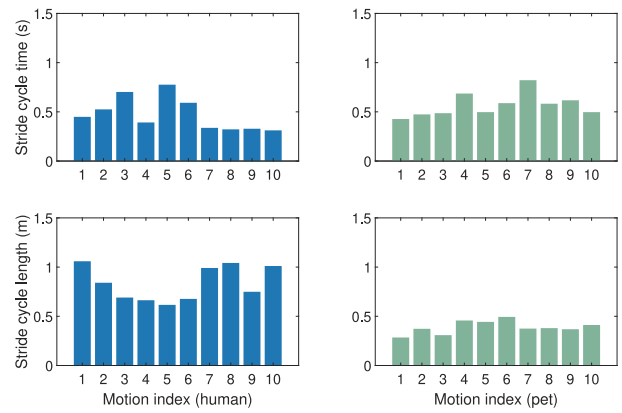


Fig. 7. Stride cycle time and stride length of human and pet.

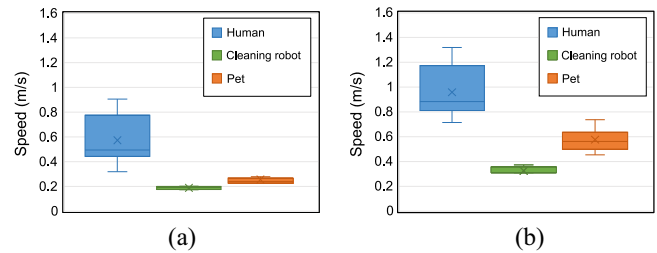


Fig. 8. (a) 25 percentile (b) 75 percentile of speed estimations for human, robot, and pet.

a particular time window. Additionally, speed deviations are taken into account to distinguish between the motions of humans, pets, and robots, as depicted in Fig. 8. Specifically, we estimate the variance, 25th percentile value, and 75th percentile value of the speed in the time window as features of speed deviations.

Vacuum cleaners, being wheeled, lack feet, and maintain relatively constant speeds, slowing only when meeting obstacles. Hence, we employ speed patterns to distinguish between vacuum robot and human movements. Specifically, the average speed and speed deviations are extracted. The average speed is derived from the mean of a target's instantaneous speeds within a specific time window. As for speed deviation, we estimate the variance, 25th and 75th percentile value of speed, as shown in Fig. 8.

2) Statistical Feature Extraction: In most cases, human and pet movements exhibit distinctive gait features. However, certain actions, such as drinking, eating, and tail wagging do not demonstrate gait characteristics. Therefore, to augment our distinction capability, we further exploit other features such as statistical properties derived from the ACF and motion statistics of CSI. These attributes prove valuable in situations where speed estimation is not reliable, such as when subjects are too distant for accurate gait estimation.

ACF Features: ACF of CSI serves as an advantageous statistic since it encapsulates a wealth of information regarding the subject's motion, while its sensitivity to location and environment is minimal. As such, we extract features based on ACF. Our observations reveal that the movements of humans, pets, and robots uniquely impact the amplitude and frequency

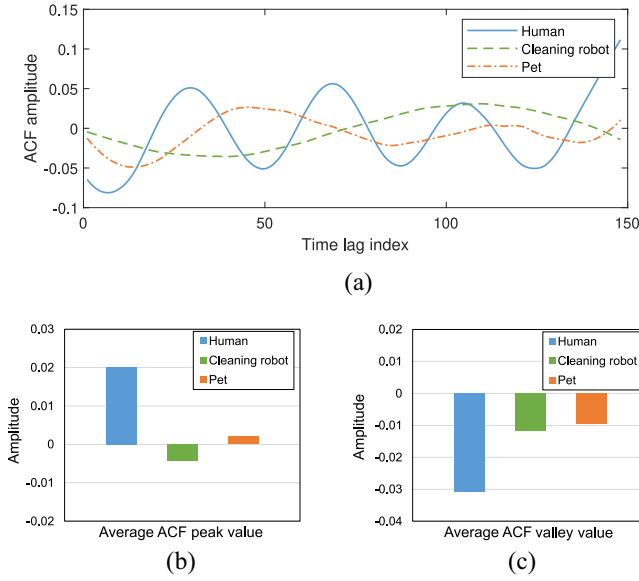


Fig. 9. (a) ACF at a time instance, (b) average ACF peak value, and (c) average ACF valley values of human, robot, and pet.

of ACF peaks at each time step. This distinction is evidenced by the variances in the amplitude and time lag of ACF peaks and valleys across different types of movements, as depicted in Fig. 9(a). To leverage these distinctions, we extract the mean values of ACF peaks and valleys separately for each movement type, as shown in Fig. 9(b) and (c). These features provide a comprehensive representation of the unique characteristics of human, pet, and robot movements and enable effective differentiation between different motion types in various environments.

Motion Statistic Features: In addition to the ACF-based features, motion statistics also provide valuable insights into the statistical characteristics of the impact of the subject's movement on the CSI. This assists in refining our ability to identify human and nonhuman movements. To capture these statistical characteristics, we compute motion statistics according to (4) and extract their mean and variance as additional features. The statistical features enhance the system's resilience to variations in the environment, making it more adaptable to real-world scenarios.

B. Recognition Model Design

SVM is a supervised learning model that utilizes associated learning algorithms to classify and perform regression analysis on data. Our SVM model uses a feature vector consisting of 11 features, including gait existence, stride length, stride cycle time, average speed, speed variance, speed 25 percentile, speed 75 percentile, ACF peaks mean, ACF valleys mean, motion statistic mean, and motion statistic variance. During the training phase, we trained SVM models employing linear kernel or Gaussian kernel, and the best model was chosen via a grid search complemented by 5-fold cross-validation. Concurrently, the features underwent standardization. The trained SVM model was subsequently tested in unknown environments, with no further training or parameter adjustments.

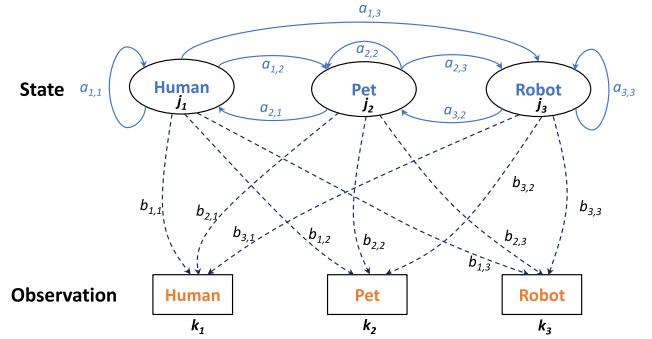


Fig. 10. Illustration of state machine.

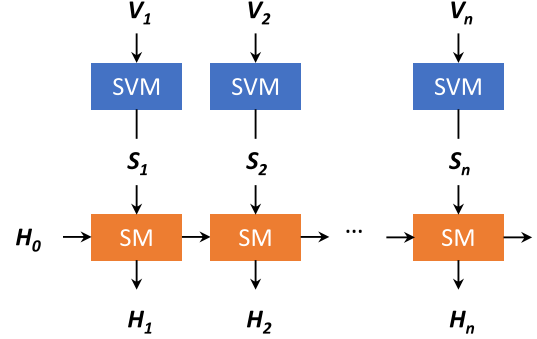


Fig. 11. Illustration of the integration of state machine and SVM classifier.

C. State Machine

In some cases, distinct continuous gait features are not available due to irregular motion during walking/running/slow moving, such as stopping, turning around, etc. Consequently, the classifier may misclassify humans as nonhuman due to the lack of obvious gait features and the presence of small motions. To address this challenge, we propose a state machine based on HMM, as shown in Fig. 10.

In the proposed state machine, we estimate the parameters using the maximum likelihood estimation method to determine the most likely probability of the observation sequence occurring. Let the emission probability $b_{j,k}$ represent the probability of observing symbol k in state j . It can be estimated as follows:

$$b_{j,k} = \frac{c_j(k)}{\sum_{k=1}^M c_j(k)} \quad (5)$$

where $c_j(k)$ denotes the number of times symbol k is observed in state j , and M represents the total number of symbols. The transition probability $a_{i,j}$ represent the probability of transitioning from state i to state j , and can also be estimated as follows:

$$a_{i,j} = \frac{c_{i,j}}{\sum_{j=1}^N c_{i,j}} \quad (6)$$

where $c_{i,j}$ is the number of transitions from state i to state j , and N denotes the total number of states.

The integration of the state machine into the system's workflow is depicted in Fig. 11. In the figure, V represents the feature vector obtained by feature extraction, S signifies the SVM-based classifier's identification outcomes, and H denotes the ultimate decision output produced by the state machine.

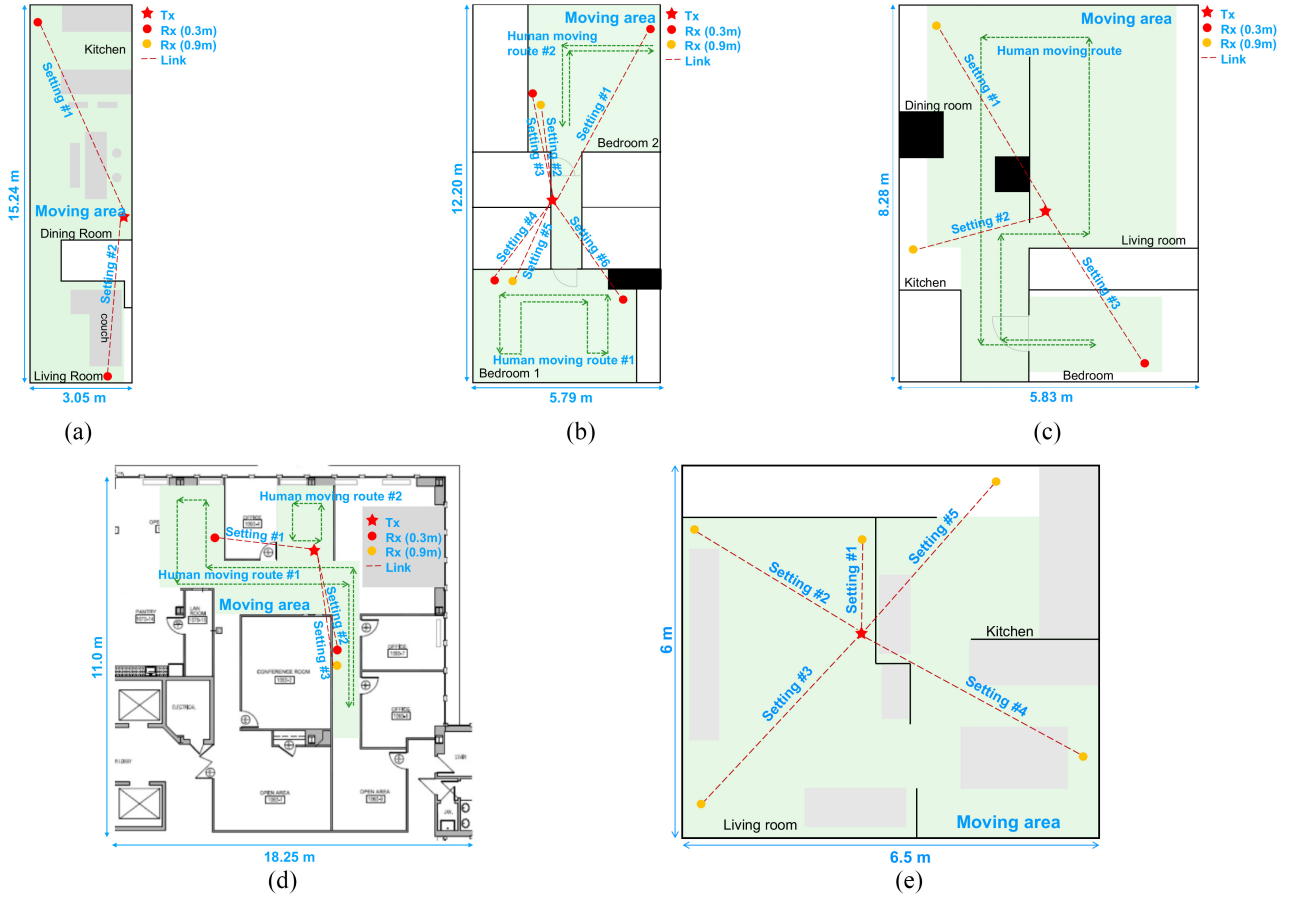


Fig. 12. Floor plans of (a) scenario I: an apartment, (b) scenario II: a townhouse, (c) scenario III: a single family house, (d) scenario IV: an office building, and (e) scenario V: an apartment. Multiple pairs of transceiver are adopted to enhance data collection efficiency and data diversity, but only data from single transceiver pair is used for feature extraction and recognition.

For each time $t = i$, the output H_i is calculated based on the Viterbi algorithm [32] and is the maximum a posteriori probability estimate of the most likely sequence of hidden true states at time $t = i$ according to the past inputs.

In Section VII, we discuss the effectiveness of HMM-based state machines in enhancing the performance of the proposed classifier and minimizing the occurrence of misclassifications due to irregular human motions.

VI. EVALUATION

This section elaborates the evaluation of WI-MoID. First, we introduce the evaluation methodologies, including experiment settings, data collection, and metrics. Then, we discuss the recognition performance from various aspects.

A. Methodology

1) *Implementation*: We have implemented the system on off-the-shelf WiFi sensing devices, as shown in Fig. 13(c). Each device has a commercial WiFi chipset which is dual-band operating on both 2.4 and 5 GHz. The main processing board is a PICO-IMX7 System-on-Module consisting of a Dual-core ARM Cortex-A7 CPU up to 1 GHz and 1-GB memory. The WiFi chipset contains two antennas with U.FL/IPX connector interfaces. The Rx receives packets transmitted from the Tx

and captures CSI with a sample rate of 1500 Hz. Both the Tx and Rx consist of two omnidirectional PCB antennas. The system runs on the 5.805-GHz channel with a bandwidth of 40 MHz.

2) *Experimental Setting*: We performed experiments in five representative indoor environments. Specifically, Scenario I corresponds to a 76.96-m² apartment, Scenario II a 70.64-m² townhouse, Scenario III a 40.27-m² single-family house, Scenario IV a 200.75-m² office building, and Scenario V a 39-m² apartment, as illustrated in Fig. 12. To obtain data from different locations and heights, we employed one transmitter and two to five receivers in every data gathering session. Receivers were positioned 4 to 10 m from the transmitter at heights varying between 0.3 and 1 m, considering the presence of walls. Note that although we used multiple transceiver pairs in each session to improve efficiency and diversity, the CSI data from different pairs are not combined. The feature extraction and recognition processes are performed based on a single-pair WiFi transceiver. In each scenario, the movement areas for humans, pets, and vacuum robots were delineated in green on Fig. 12.

3) *Data Collection*: We gathered motion data from five human subjects (three males and two females) with heights ranging from 155 to 194 cm, alongside two dogs with lengths ranging from 40 cm to 1 m and a 17-lb cat. The cleaning

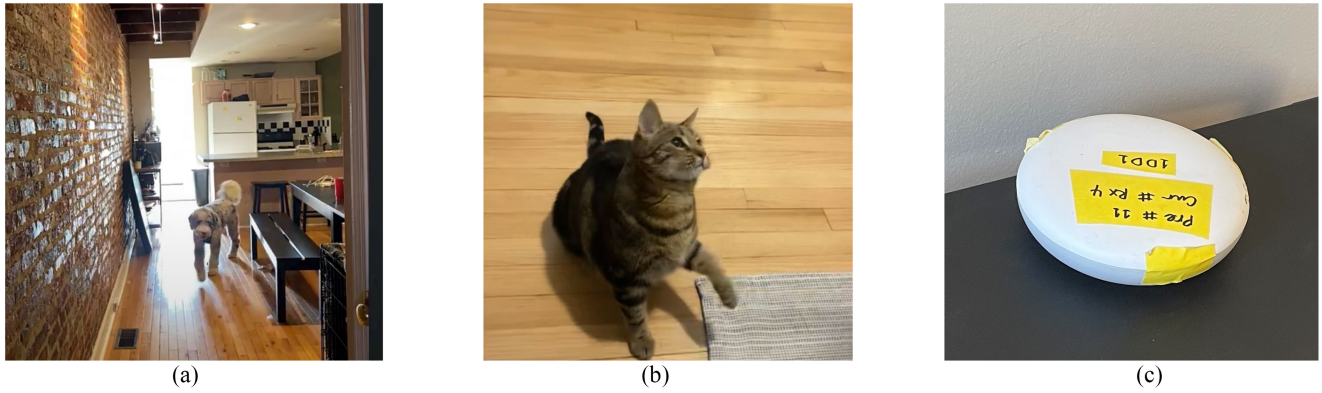


Fig. 13. (a) Dog in Scenario I and (b) cat in Scenario V, and (c) off-the-shelf WiFi edge device.

robot data is collected with an iRobot i3 vacuum cleaner. All data collection was conducted in a natural manner without any restrictions. Specifically, the human movement data was captured while individuals freely walked around the environments, with the freedom to roam and stop at will. They engaged in various activities, such as using mobile phones or playing games. Continuous data collection was performed for the pets' movement throughout the day as it walked around, ate, drank, and wagged its tail as usual. Fig. 13(a) and (b) show the pets in Scenarios I and V, respectively. We gathered the vacuum cleaner's motion data when it was cleaning the floors furnished with items like tables and chairs, with the robot moving freely within the accessible spaces to perform its task.

We conducted the experiments over a period of seven months, where the subjects initiated their movements from arbitrary locations. Movement durations ranged from 2 s to 20 min during each data collection session. In total, we accumulated approximately 369 min of human motion data, 225 min of pet motion data, and 100 min of robot motion data. We applied motion statistics to detect motion and segmented the data into 6-s instances. This yielded 3202 instances of human motion data, 631 instances of pet motion data, and 823 instances of robot motion data. For the training and validation process, we randomly allocated 90% of the data from Scenarios I and II to form the training data set, reserving the remaining 10% for validation. We used Scenarios III, IV, and V data exclusively to test the model's resilience to unknown environments, excluding them from the training or validation phases.

To evaluate the model's adaptability to a variety of human motion types, we gathered 356 instances of sneaking, 367 instances of running data, and 215 instances of intrusion data in Scenarios II and III. During intrusion data collection, the participants mimic intruders by performing a series of continuous motions, including sneaking, crawling, walking, and running randomly in the environment. Additionally, 306 instances of ceiling fan motion were collected in Scenarios I and III to evaluate the model's versatility in recognizing various nonhuman subject movements. The diverse human and fan motion data, collected in Scenarios I and II, were incorporated into our training and validation data sets. Conversely, data from Scenario III were leveraged to assess the model's performance

in an unknown environment, offering a comprehensive evaluation of its ability to generalize and adapt.

4) *Evaluation Metrics:* The effectiveness of W1-MOID is gauged through the following key metrics.

- 1) Recall or true positive rate (TPR) represents the likelihood of the system correctly identifying the target.
- 2) Precision, alternatively termed as positive predictive value (PPV), signifies the proportion of accurate recognition results for a specific class.
- 3) Accuracy is the ratio of correctly identified data to the total data pool.
- 4) False alarm or false positive rate (FPR) measures the probability of the system misidentifying the target's motion.

The system's performance is considered improved when there is an increase in accuracy, precision, and recall, along with a decrease in the false alarm rate. These metrics collectively serve as a measure of the system's efficacy.

B. Recognition Performance

To evaluate the effectiveness of W1-MOID, we conducted experiments using data collected from Scenarios I and II as the training and validation data set. First, we extracted features from the data and trained SVM models with different kernels. To ensure the model's ability to generalize, we incorporated a 5-fold cross-validation strategy to randomize the training and validation data set. The model that demonstrated the highest validation accuracy was chosen as the final model. The optimal SVM model's performance was gauged by its accuracy, precision, recall, and false alarm rate, the details of which are presented in Table I. The table shows that W1-MOID achieved high validation accuracy of 97.34%, paired with a minimal false alarm rate of 1.75%, indicating its ability to accurately identify the motion of human and nonhuman subjects.

1) *Adaptivity to Environment Changes:* To assess the generalizability of the trained model across new contexts, we employed a testing data set gathered from previously unobserved settings (Scenarios III and IV). The model's testing accuracy was ascertained without further retraining is documented in Table I. We observed that our pretrained model

TABLE I
HUMAN AND NONHUMAN MOTION IDENTIFICATION PERFORMANCE

	Overall Accuracy	Overall Recall	Overall Precision	Overall False Alarm	Human			iRobot			Pet		
					Recall	Precision	False Alarm	Recall	Precision	False Alarm	Recall	Precision	False Alarm
Validation	97.34	96.84	93.21	1.75	98.31	99.96	3.33	94.12	99.21	1.69	98.08	80.45	0.23
Testing	95.98	92.90	97.43	1.99	99.52	95.51	1.59	97.01	100.00	0.19	82.19	96.77	4.18

All the numbers in this table are percentages.

TABLE II
ADAPTIVITY TO OTHER NONHUMAN SUBJECTS

	Validation Accuracy(%)	Testing Accuracy(%)	False Alarm(%)
Human, iRobot, Pet	97.34	95.98	1.75
Human, iRobot, Pet, Fan	97.58	94.13	1.48

achieved a high testing accuracy and precision of 95.98% and 93.93%, respectively, along with a low false alarm rate of 3.66%. These results demonstrate the robustness of WI-MOID in recognizing humans, pets, and robots through walls in new environments. Furthermore, they confirm that our trained model is resilient to environmental changes and independent of the training environment. This attribute facilitates quick and effortless deployment in new contexts, obviating the need for additional user exertion.

2) *Adaptivity to Other Nonhuman Subject*: While pets and robots are typically the primary sources of nonhuman motion in indoor scenarios, electrical appliances such as fans can also introduce motion. Therefore, it is necessary to evaluate the robustness of our algorithms in recognizing motion caused by electrical appliances. In particular, we assessed the robustness of WI-MOID to fan motions. To achieve this, we gathered 306 instances of CSI data while the fan was operating in the environment. The results of validation accuracy, testing accuracy, and false alarm rate are presented in Table II. Even with the inclusion of fan motion, the validation and testing accuracies remain high, and the false alarm rate is low. These results indicate that our model is adaptable to other electrical appliances' motion, such as fan motion, due to the effectiveness of the proposed features. Our approach encompasses the extraction of attributes that carry physical significance and simultaneously encapsulates statistical elements. These elements effectively mirror the influences of both human and nonhuman movements on the CSI within the environment. Since electrical appliances typically operate continuously and periodically, they affect the statistics differently from human motions, allowing our model to robustly distinguish electrical appliances' motion from human motion.

3) *Adaptivity to Human Motion Types*: To gauge our model's flexibility in handling other human motion types, we incorporated sneaking and running activities into our experimental setup. A total of 106 min of CSI data encapsulating these motions was collected, allowing us to extract 723 6-s instances representing various human movements. We trained and validated the SVM model using various human motion data from Scenarios I and II. The model's robustness was further assessed by testing it with data from Scenarios III and IV, which were not part of the initial training set. All data

TABLE III
STATE MACHINE PERFORMANCE EVALUATION

		Without State Machine	With State Machine	Improvement
Human	Accuracy(%)	98.29	99.96	1.67
	False Alarm(%)	4.49	3.33	1.16
iRobot	Accuracy(%)	97.36	99.21	1.85
	False Alarm(%)	4.25	1.69	2.56
Pet	Accuracy(%)	62.47	80.45	17.98
	False Alarm(%)	2.2	0.23	1.97

subsets utilized for training, validation, and testing comprised an all-encompassing array of human movements, inclusive of walking, running, and sneaking. The results, illustrated in Fig. 14, revealed a high recognition accuracy of 92.55% even with the inclusion of more diverse human motion types. This underscores the adaptability of WI-MOID to accommodate a variety of human movements.

VII. DISCUSSION

A. Effectiveness of State Machine

To evaluate the effectiveness of the proposed state machine, we compared the recognition accuracy of WI-MOID with and without the state machine. Table III presents the accuracy and false alarm rates for human, robot, and pet recognition. The results demonstrate that the state machine improves the recognition accuracy of human, robot, and pet by 1.67%, 1.85%, and 17.98%, respectively, compared to the system without the state machine. The state machine improves the accuracy of pet recognition significantly by addressing the challenges posed by pet motion, which are difficult for SVM classifiers to handle. Specifically, pets can be misclassified as humans due to their similar gait patterns, and they can also be mistaken for cleaning robots when they move slowly or wave their tails, generating small motions that can interfere with CSI signals in a manner similar to cleaning robots. In addition, the proposed state machine reduces the false alarm rates of human, robot, and pet recognition by 1.16%, 2.56%, and 1.97%, respectively. The reduction in false alarms and the increase in human detection accuracy are particularly noteworthy since the state machine helps avoid misclassifying cases where human motion lacks an obvious gait pattern. These results demonstrate that the state machine effectively improves recognition accuracy and reduces the false alarm rate, enhancing the robustness and reliability of WI-MOID.

B. Latency

We assessed the computational performance of WI-MOID by evaluating the time taken by model training on the



Fig. 14. Adaptivity to human motion types.

TABLE IV
RECOGNITION PERFORMANCE WITH FEATURE SELECTION

	Overall Accuracy(%)	Average Recall(%)	Average Precision(%)
All Features	97.34	96.84	93.21
Feature selection	94.52	91.79	86.19

desktop and real-time processing on the edge device. For SVM classifier training, we run the MATLAB code on a desktop computer equipped with an Intel Core i7-8700K CPU at 3.70 GHz and 16-GB RAM. The training and testing of the SVM classifier with 1000 instances take 1.084 and 0.019 s, respectively. Since the feature vectors are environment independent, the training process can be done offline, and the time required for training is negligible. During real-time deployment on edge devices, the Rx keeps capturing CSI and running the WI-MOID algorithm to output identification decisions every other second. The total processing time for an instance is 0.043 s, where the feature extraction takes 0.027 s, and classification takes 0.016 s. This evaluation is derived from the average results of a 20-min test.

C. Feature Efficiency

To evaluate the effectiveness of the features designed in our model, we conducted a comparative analysis utilizing a model trained with a 95% principal component analysis (PCA)-driven approach, which was designed to decrease the feature vector's dimensionality. This strategy preserved only the principal components accounting for 95% of the total variance. The SVM model underwent retraining with the condensed feature set, using the identical data set. The comparison results are recorded in Table IV, which demonstrates that the accuracy of the PCA-trained model was lower than that of the original model that utilized our designed features. This observation indicates the effectiveness of the designed features, which contributes to the high accuracy of our model.

D. Length of Motion Segments

We investigated the impact of the length of motion segments on the performance of WI-MOID, as shown in Fig. 15. Our results indicate that the length of motion segments does not have a significant effect on the accuracy of the system. While shorter motion segments may provide more specific information about the motion, longer motion segments may offer a broader perspective of the overall motion. However,

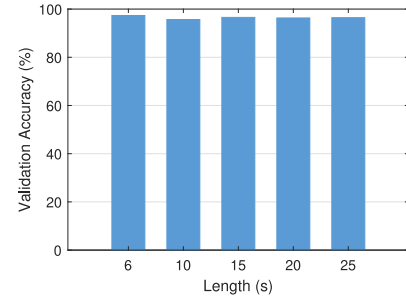


Fig. 15. Impact of length of motion segments.

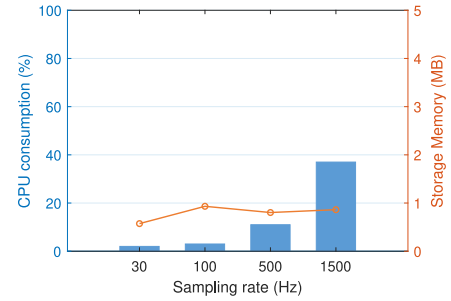


Fig. 16. CPU and memory requirements.

TABLE V
PERFORMANCE AT DIFFERENCE SOUNDING RATES

sounding rates (Hz)	1500	500	100	30
Accuracy (%)	97.34	94.36	94.31	91.8
Recall (%)	96.84	94.57	95.43	91.94
Precision (%)	93.21	88.48	87.97	84.16
False alarm (%)	1.75	3.06	3.38	4.68
Accuracy w/o state machine (%)	93.58	88.48	85.88	86.03

our findings suggest that neither length leads to a significant difference in the accuracy of the system. Therefore, we conclude that WI-MOID is robust to variations in the length of motion segments, ranging from 6 to 25 s, and can accurately analyze motions of different durations.

E. Sounding Rate

In this study, we also explore the impact of different sounding frequencies on performance. We compare the detection rate and false alarm rate of WI-MOID at four different sounding frequencies: 1) 1500; 2) 500; 3) 100; and 4) 30 Hz, as shown in Table V. The results demonstrate that the proposed algorithm achieves high accuracy and low false alarm rates across a range of sounding rates, with accuracy improving as the sounding rate increases. Notably, we also observed that the algorithm's accuracy remains satisfactory even at low sounding rates, indicating its robustness to variations in data acquisition frequency. This suggests that the algorithm can be effectively used in resource-constrained scenarios where high sounding rates may not be feasible due to memory or power limitations, without compromising its accuracy. Therefore, the proposed algorithm offers a scalable solution for real-time data processing applications on resource-constrained edge devices.

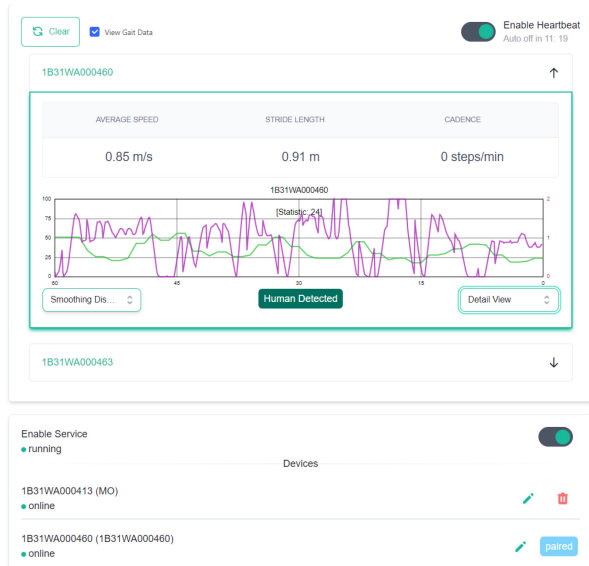


Fig. 17. Web GUI of WI-MOID.

F. Edge Device Deployment

The computational efficiency and storage requirements of the proposed algorithm were evaluated on commercialized edge devices with Marvell chipset, as shown in Fig. 16. Our results indicate that the algorithm is highly optimized for edge computing, with minimal CPU consumption and storage usage. This makes it an ideal candidate for deployment on resource-constrained edge devices. Moreover, the algorithm's scalability enables it to adapt to varying requirements and device specifications, making it a versatile solution for real-time data processing at the edge. To display the identification results, we develop a Web GUI, as shown in Fig. 17, which visualizes the live motion statistics, speed estimations, gait features, and the WI-MOID identification results.

VIII. CONCLUSION

In this study, we introduce an unprecedented system that identifies the movements of various human and nonhuman subjects utilizing readily available WiFi devices. The system identifies moving subjects by analyzing both physically interpretable and statistical attributes of the movement. A comprehensive assessment in diverse indoor scenarios validates the resilience of WI-MOID in the real-world against various nonhuman subjects and changing conditions. WI-MOID has been implemented on edge devices and is ready to be deployed ubiquitously for extensive use.

REFERENCES

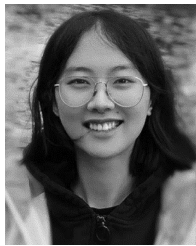
- [1] K. J. R. Liu and B. Wang, *Wireless AI: Wireless Sensing, Positioning, IoT, and Communications*. Cambridge, U.K.: Cambridge Univ. Press, 2019.
- [2] 2023–2024 APPA National Pet Owners Survey, American Pet Products Association, Stamford, CT, USA, 2023.
- [3] J. L. Kernek et al., "Radar sensing in assisted living: An overview," in *Proc. IEEE MTT-S Int. Microw. Biomed. Conf. (IMBioC)*, vol. 1, 2019, pp. 1–4.
- [4] N. H. Tan, R. Y. Wong, A. Desjardins, S. A. Munson, and J. Pierce, "Monitoring pets, deterring intruders, and casually spying on neighbors: Everyday uses of smart home cameras," in *Proc. CHI Conf. Human Factors Comput. Syst.*, 2022, pp. 1–25.
- [5] A. Naser, A. Lotfi, J. Zhong, and J. He, "Human activity of daily living recognition in presence of an animal pet using thermal sensor array," in *Proc. 13th ACM Int. Conf. Pervasive Technol. Related Assist. Environ.*, pp. 1–6.
- [6] J. Gong, X. Zhang, J. Ren, and Y. Zhang, "The invisible shadow: How security cameras leak private activities," in *Proc. ACM SIGSAC Conf. Comput. Commun. Security*, 2021, pp. 2780–2793.
- [7] A. Bhattacharya and R. Vaughan, "Deep learning radar design for breathing and fall detection," *IEEE Sensors J.*, vol. 20, no. 9, pp. 5072–5085, May 2020.
- [8] E. Cardillo, C. Li, and A. Caddemi, "Vital sign detection and radar self-motion cancellation through clutter identification," *IEEE Trans. Microw. Theory Tech.*, vol. 69, pp. 1932–1942, Mar. 2021.
- [9] Y. Lin, Y. Gao, B. Li, and W. Dong, "Revisiting indoor intrusion detection with WiFi signals: Do not panic over a pet!" *IEEE Internet Things J.*, vol. 7, no. 10, pp. 10437–10449, Oct. 2020.
- [10] F. Zhang, C. Chen, B. Wang, and K. J. R. Liu, "WiSpeed: A statistical electromagnetic approach for device-free indoor speed estimation," *IEEE Internet Things J.*, vol. 5, no. 3, pp. 2163–2177, Jun. 2018.
- [11] E. Soltanaghaei et al., "Robust and practical WiFi human sensing using on-device learning with a domain adaptive model," in *Proc. 7th ACM Int. Conf. Syst. Energy-Efficient Build., Cities, Transp.*, 2020, pp. 150–159.
- [12] P. Wang et al., "Method for distinguishing humans and animals in vital signs monitoring using IR-UWB radar," *Int. J. Environ. Res. Public Health*, vol. 16, no. 22, p. 4462, 2019.
- [13] M. Forouzanfar, M. Mabrouk, S. Rajan, M. Bolic, H. R. Dajani, and V. Z. Groza, "Event recognition for contactless activity monitoring using phase-modulated continuous wave radar," *IEEE Trans. Biomed. Eng.*, vol. 64, no. 2, pp. 479–491, Feb. 2017.
- [14] C. Wu, B. Wang, O. C. Au, and K. J. R. Liu, "WiFi can do more: Toward ubiquitous wireless sensing," *IEEE Commun. Stand. Mag.*, vol. 6, no. 2, pp. 42–49, Jun. 2022.
- [15] B. Wang, Q. Xu, C. Chen, F. Zhang, and K. J. R. Liu, "The promise of radio analytics: A future paradigm of wireless positioning, tracking, and sensing," *IEEE Signal Process. Mag.*, vol. 35, no. 3, pp. 59–80, May 2018.
- [16] F. Zhang, C. Wu, B. Wang, H.-Q. Lai, Y. Han, and K. J. R. Liu, "WiDetect: Robust motion detection with a statistical electromagnetic model," *Proc. ACM Interact., Mobile, Wearable Ubiquitous Technol.*, vol. 3, no. 3, pp. 1–24, 2019.
- [17] D. Wu, D. Zhang, C. Xu, H. Wang, and X. Li, "Device-free WiFi human sensing: From pattern-based to model-based approaches," *IEEE Commun. Mag.*, vol. 55, no. 10, pp. 91–97, Oct. 2017.
- [18] F. Wang, W. Gong, and J. Liu, "On spatial diversity in WiFi-based human activity recognition: A deep learning-based approach," *IEEE Internet Things J.*, vol. 6, no. 2, pp. 2035–2047, Apr. 2019.
- [19] J. Wang, Y. Zhao, X. Ma, Q. Gao, M. Pan, and H. Wang, "Cross-scenario device-free activity recognition based on deep adversarial networks," *IEEE Trans. Veh. Technol.*, vol. 69, no. 5, pp. 5416–5425, May 2020.
- [20] F. Wang, W. Gong, J. Liu, and K. Wu, "Channel selective activity recognition with WiFi: A deep learning approach exploring wideband information," *IEEE Trans. Netw. Sci. Eng.*, vol. 7, no. 1, pp. 181–192, Jan.–Mar. 2020.
- [21] Y. Ma et al., "Location- and person-independent activity recognition with WiFi, deep neural networks, and reinforcement learning," *ACM Trans. Internet Things*, vol. 2, no. 1, p. 3, 2021.
- [22] D. Wang, J. Yang, W. Cui, L. Xie, and S. Sun, "Multimodal CSI-based human activity recognition using GANs," *IEEE Internet Things J.*, vol. 8, no. 24, pp. 17345–17355, Dec. 2021.
- [23] C. Wu, F. Zhang, Y. Hu, and K. J. R. Liu, "GaitWay: Monitoring and recognizing gait speed through the walls," *IEEE Trans. Mobile Comput.*, vol. 20, no. 6, pp. 2186–2199, Jun. 2021.
- [24] Y. Chen, W. Dong, Y. Gao, X. Liu, and T. Gu, "Rapid: A multimodal and device-free approach using noise estimation for robust person identification," *Proc. ACM Interact. Mobile Wearable Ubiquitous Technol.*, vol. 1, no. 3, p. 41, 2017.
- [25] Y. Cao, Z. Zhou, C. Zhu, P. Duan, X. Chen, and J. Li, "A lightweight deep learning algorithm for WiFi-based identity recognition," *IEEE Internet Things J.*, vol. 8, no. 24, pp. 17449–17459, Dec. 2021.
- [26] Y. Xu, W. Yang, M. Chen, S. Chen, and L. Huang, "Attention-based gait recognition and walking direction estimation in WiFi networks," *IEEE Trans. Mobile Comput.*, vol. 21, no. 2, pp. 465–479, Feb. 2022.

- [27] C. Wu, F. Zhang, B. Wang, and K. J. R. Liu, "EasiTrack: Decimeter-level indoor tracking with graph-based particle filtering," *IEEE Internet Things J.*, vol. 7, no. 3, pp. 2397–2411, Mar. 2020.
- [28] Y. Zheng et al., "Zero-effort cross-domain gesture recognition with WiFi," in *Proc. 17th Annu. Int. Conf. Mobile Syst., Appl., Services*, 2019, pp. 313–325.
- [29] G. Zhu, C. Wu, B. Wang, and K. J. R. Liu, "Floor plan reconstruction with high-precision RF-based tracking," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, 2022, pp. 5073–5077.
- [30] G. Zhu, C. Wu, B. Wang, and K. J. R. Liu, "EZMap: Boosting automatic floor plan construction with high-precision robotic tracking," *IEEE Internet Things J.*, vol. 10, no. 8, pp. 6988–6998, Apr. 2023.
- [31] G. Zhu, C. Wu, X. Zeng, B. Wang, and K. J. R. Liu, "Who moved my cheese? Human and non-human motion recognition with WiFi," in *Proc. IEEE 19th Int. Conf. Mobile Ad Hoc Smart Syst. (MASS)*, 2022, pp. 476–484.
- [32] A. Viterbi, "Error bounds for convolutional codes and an asymptotically optimum decoding algorithm," *IEEE Trans. Inf. Theory*, vol. 13, no. 2, pp. 260–269, Apr. 1967.



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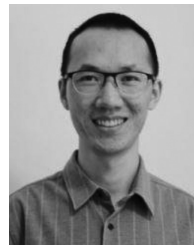


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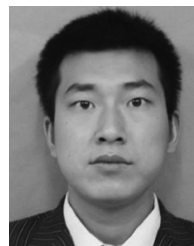
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